



HK IMO Training 2021-2022
Problem Set 14
For 15 January 2022



Problem 53. Let AB be a chord of a circle Γ . Let M be the midpoint of AB . The circle Ω with centre A and passing through M meets Γ at C and D . Let N be the midpoint of the major arc AB in Γ . The segment AN meets CD at P . The line MP meets Ω again at Q . Prove that CD is perpendicular to NQ .

Problem 54. Let n be a positive integer. Suppose $\{x_k\}$ is a sequence of integers such that $0 \leq x_j \leq n$ for all j . Define another sequence $\{y_k\}$ by

$$y_k = \sqrt[k]{x_1^k + x_2^k + \cdots + x_n^k}.$$

If there are infinitely many terms in $\{y_k\}$ which are integers, prove that all the terms in $\{y_k\}$ are integers.

Problem 55. There is a 2022×2022 grid where the grid lines are white in colour. We draw n rectangles in red colour using some of the grid lines as sides. It turns out that all the grid lines of the grid become red. Find the smallest possible value of n .

Problem 56. Let $\{a_n\}$ be a sequence of positive integers such that $\gcd(a_m, a_n) = a_{\gcd(m, n)}$ for any positive integers m and n . Prove that $\prod_{d|n} a_d^{\mu(\frac{n}{d})}$ is an integer for every positive integer n , where the product runs through all positive divisors d of n . (Here, $\mu(m)$ is the Möbius function.)